EXPERIMENT – 6

Snort IDS/IPS to Detect and Prevent Real-Time Threats Using TryHackMe Platform

Aim

To understand and demonstrate the working of Snort as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) by detecting real-time network threats using the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Snort IDS/IPS Tool (Pre-installed in AttackBox)

Theory

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are essential components of network security used to monitor, detect, and prevent malicious activities.
Snort is an open-source, signature-based IDS/IPS capable of real-time traffic analysis and packet inspection.
TryHackMe provides a safe and legal environment to practice IDS/IPS concepts by simulating real-world attacks and defenses without impacting live systems.

Procedure

Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using registered credentials.

Step 2: Open Snort Room

1. In the search bar, type **Snort**.
2. Select a room related to **Snort IDS/IPS**.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Verify Snort Installation

1. In the terminal, check Snort version:
2. snort --version
3. Ensure Snort is successfully installed.

Step 5: Run Snort in IDS Mode

1. Start Snort in console mode:
2. sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
3. Snort begins monitoring network traffic in real time.

Step 6: Detect Network Threats

1. Perform network scanning or simulated attacks as instructed in the TryHackMe room.
2. Observe alerts generated by Snort.
3. Analyze alerts indicating suspicious or malicious activity.

Step 7: Configure Snort as IPS (Optional)

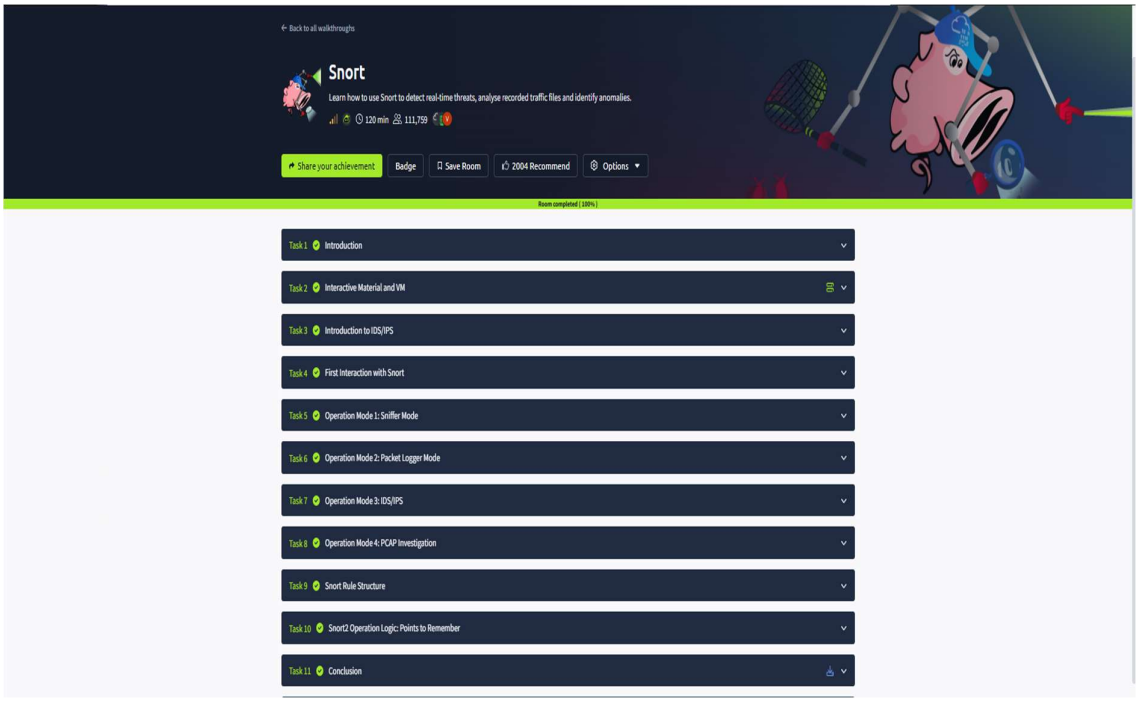
1. Run Snort in inline (IPS) mode:
2. sudo snort -Q --daq afpacket -c /etc/snort/snort.conf -i eth0
3. Configure rules to drop malicious packets.
4. Observe blocked traffic in real time.

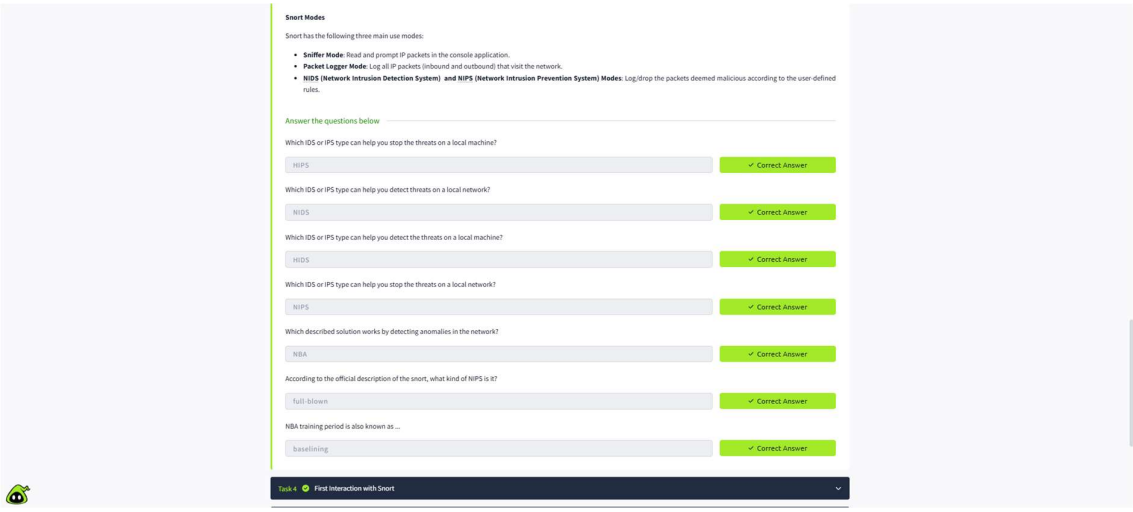
Step 8: Analyze Alerts

1. Review generated alerts on the terminal.
2. Identify attack types such as port scanning, brute force, or suspicious packets.

Output

- Snort successfully detects malicious traffic.
- Real-time alerts are displayed in the terminal.
- Screenshots of alerts and completed tasks are recorded.





Snort Modes

Snort has the following three main use modes:

- **Sniffer Mode**: Read and prompt IP packets in the console application.
- **Packet Logger Mode**: Log all IP packets (inbound and outbound) that visit the network.
- **NIDS (Network Intrusion Detection System) and NIPS (Network Intrusion Prevention System) Modes**: Log/drop the packets deemed malicious according to the user-defined rules.

Answer the questions below

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS — ✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS — ✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS — ✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS — ✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA — ✓ Correct Answer

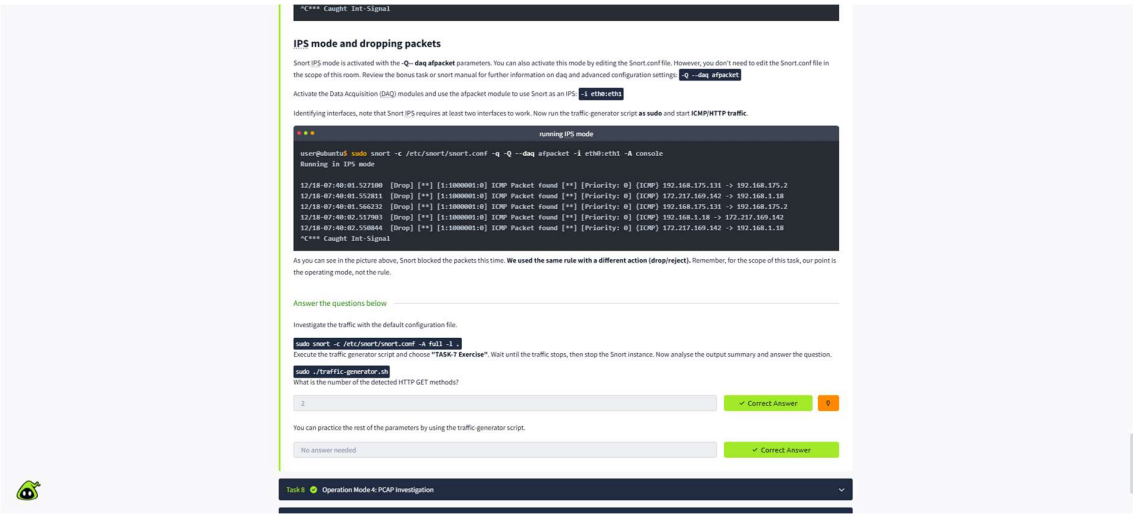
According to the official description of the snort, what kind of NIPS is it?

full-blown — ✓ Correct Answer

NBA training period is also known as ...

baselining — ✓ Correct Answer

Task 4 First Interaction with Snort



```
^C*** Caught Int-Signal
```

IPS mode and dropping packets

Snort IPS mode is activated with the **-Q-- daq afpacket** parameters. You can also activate this mode by editing the Snort.conf file. However, you don't need to edit the Snort.conf file in the scope of this room. Review the bonus task or snort manual for further information on daq and advanced configuration settings: `-Q --daq afpacket`

Activate the Data Acquisition (DAQ) modules and use the afpacket module to use Snort as an IPS: `-i eth0:eth1`

Identifying interfaces, note that Snort IPS requires at least two interfaces to work. Now run the traffic-generator script **as sudo** and start **ICMP/HTTP traffic**.

running IPS mode

```
user@ubuntu$ sudo snort -c /etc/snort/snort.conf -q -Q --daq afpacket -i eth0:eth1 -A console
Running in IPS mode

12/18-07:40:01.527100  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.131 -> 192.168.175.2
12/18-07:40:01.552811  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 172.217.169.142 -> 192.168.1.18
12/18-07:40:01.566232  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.131 -> 192.168.175.2
12/18-07:40:02.517903  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.1.18 -> 172.217.169.142
12/18-07:40:02.550844  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 172.217.169.142 -> 192.168.1.18
^C*** Caught Int-Signal
```

As you can see in the picture above, Snort blocked the packets this time. **We used the same rule with a different action (drop/reject).** Remember, for the scope of this task, our point is the operating mode, not the rule.

Answer the questions below

Investigate the traffic with the default configuration file.

`sudo snort -c /etc/snort/snort.conf -A full -l .`

Execute the traffic generator script and choose "**TASK-7 Exercise**". Wait until the traffic stops, then stop the Snort instance. Now analyse the output summary and answer the question.

`sudo ./traffic-generator.sh`

What is the number of the detected HTTP GET methods?

2 — ✓ Correct Answer

You can practice the rest of the parameters by using the traffic-generator script.

No answer needed — ✓ Correct Answer

Task 8 Operation Mode 4: PCAP Investigation

Result

Thus, Snort IDS/IPS was successfully used to detect and prevent real-time network threats using the TryHackMe platform.

The experiment enhanced understanding of intrusion detection, prevention mechanisms, and real-time network security monitoring.